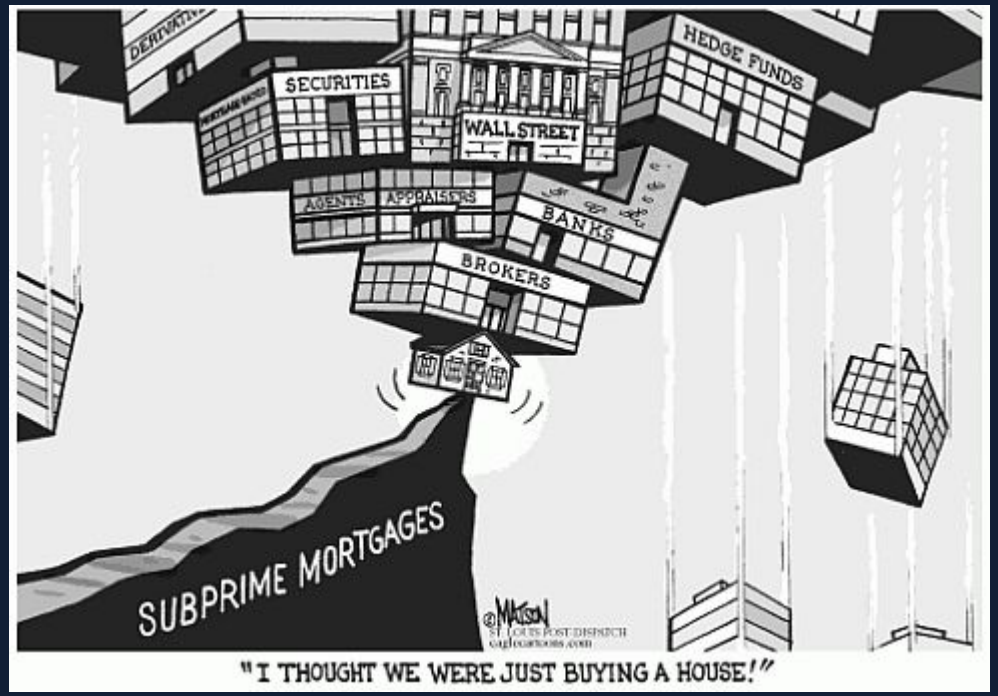
Government debt and Economic Recovery

A Cross-Country Analysis of the
2007/08 Subprime Crisis

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October 21, 2026



"I THOUGHT WE WERE JUST BUYING A HOUSE!"

Research Question & Objectives

Research Question:
How is the change in government debt associated with economic recovery following the 2008 financial crisis?

- ➔ **for this use two-sided t-tests to assess the statistical significance of the OLS coefficients**
- ➔ **H0 Hypothesis: There is no linear relationship between changes in government debt and GDP growth.**
- ➔ **H1: There is a linear relationship between changes in government debt and GDP growth**

- ✓ **Analyze 20 selected countries**
- ✓ **Compare the crisis period (2007 - 2010) and recovery period (2010 - 2015)**
- ✓ **Examine relationships using correlation and regression analysis**

Countries Included in the Analysis

19 selected European countries and the United States

Austria	Lithuania
Belgium	Luxembourg
Estonia	Netherlands
Finland	Portugal
France	Slovakia
Germany	Slovenia
Greece	Spain
Ireland	Switzerland
Italy	United Kingdom
Latvia	United States

Statistical Methods Explained

Three approaches to analyzing government debt and GDP growth



Pearson-Correlation

- it measures how strong 2 variables are linear correlated to each other
- +1 means perfect positive correlation
- -1 means perfect negative correlation
- in our example: How strong is rising the debt correlate to economic growth in the years 2007 - 2010 and 2010 - 2015



Linear Regression

- estimates how GDP growth is statistically
- how does a dependent variable (Y) change once an independent variable (X) is altered?
- in this case: how is GDP growth associated with changes in government debt?
- estimates the statistical relationship between government debt and GDP growth

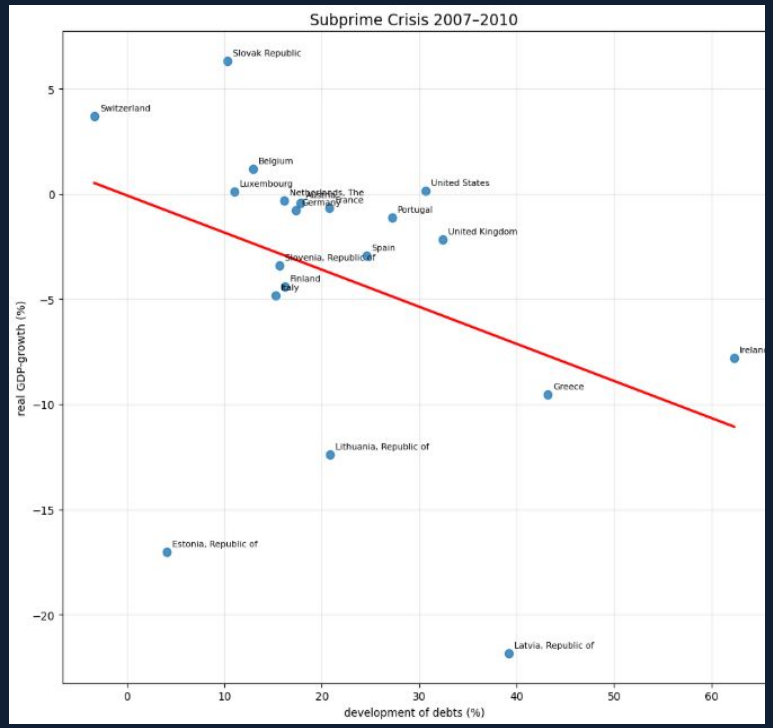


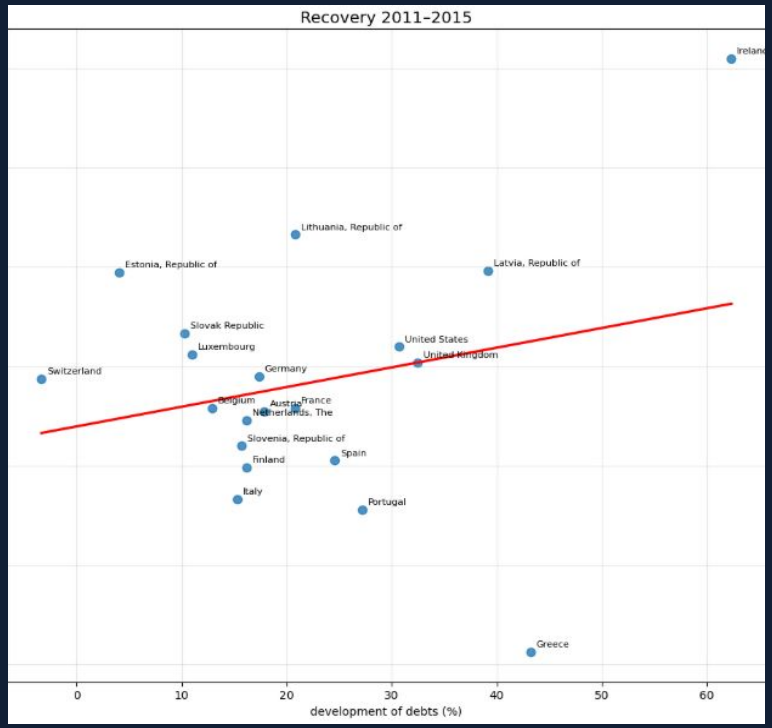
Multiple Regression

- examines the relationship between one dependent variable and multiple independent variables
- dependent variable: GDP growth
- independent variables:
  - government debts
  - inflation
  - unemployment

Statistical Analysis & Key Findings

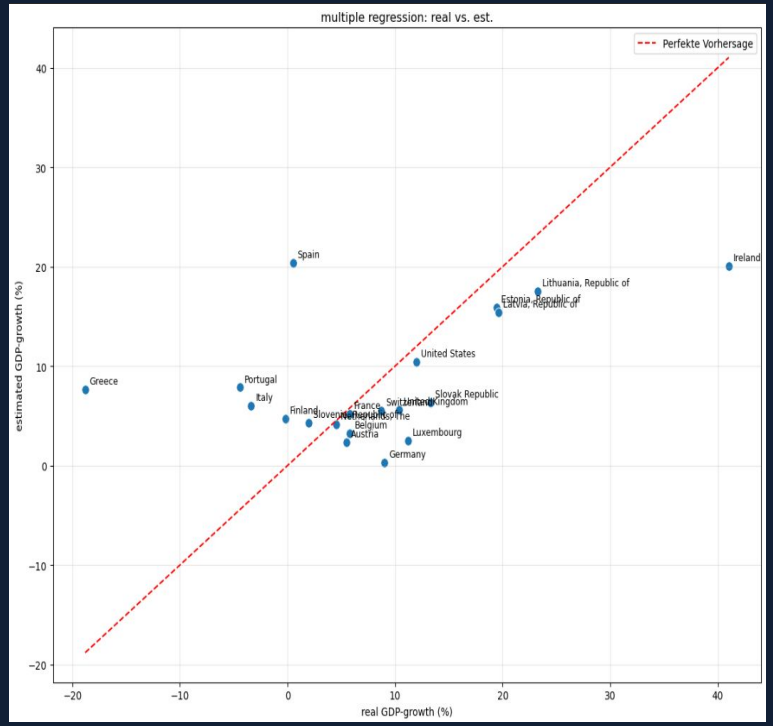
Pearson Correlation & OLS regression - 20 selected countries





Statistical Analysis & Key Findings

Multiple Regression - 20 selected countries



Statistical Analysis & Key Findings

Interpretation

, *Metric*	*Subprime-crisis 2007 - 2010*	*Recovery 2010 - 2015*	*interpretation*
Pearson-coefficient (r)	**-0.3775**	**+0.2381**	**A moderate negative correlation was observed between changes in the government debt-to-GDP ratio and GDP growth during the crisis.** **However, could increased government borrowing have mitigated the economic downturn? Would GDP growth have been even lower without additional borrowing?** **Our correlation analysis cannot answer these counterfactual questions.**

Statistical Analysis & Key Findings

Interpretation

, *Metric*	*Subprime-crisis 2007 - 2010*	*Recovery 2010 - 2015*	*interpretation*
OLS-Regression	**-0.1764**	**+0.1985**	**OLS regression shows the same direction of association as Pearson correlation: negative during the crisis and positive during the recovery period**
p-Value	**0.101**	**0.312**	**H_0 ($\beta = 0$) cannot be rejected at the 5% significance level.**
R^2	**0.142**	**0.057**	**The model explains 14.2% of the variation in GDP growth for Subprime-crisis period. 5.7% of the variation in GDP growth is explained by the model for recovery-period**

Economic Interpretation

background and causal effects

- **No statistically significant relationship**

- **Hypothesis: Additional government borrowing might have mitigated the economic downturn. Without it, GDP could have fallen further**

- **structural differences - how robust and safe have countries been in 2007?**

- **Indicators:**
  - **government debt level in 2007 before the crisis spread**
  - **level of private debts**
  - **how are the economies of countries designed (stable or volatile)**
  - **how many and what kind of assets the country has (e.g. natural resources like oil a.s.o.)**

- **Conclusion: Correlation does not imply causation**

Data Sources

- ➔ *International Monetary Fund (IMF)*
- ➔ *World Bank*